MINSEOK JOO

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RESEARCH INTERESTS

Generative and geometric models for understanding the physical world, with a focus on **video world models**, **video generation**, and **3D vision**.

EDUCATION

Korea University — Seoul, Republic of Korea
Integrated M.S.–Ph.D. Program in Computer Science and Engineering — Mar 2022–Present
Advisor: Prof. Hyunwoo J. Kim (KAIST)

Korea University — Seoul, Republic of Korea
B.S. in Biomedical Engineering — Mar 2016–Feb 2022
Second Major: Artificial Intelligence

RESEARCH EXPERIENCE

Machine Learning & Vision Lab — Feb 2022–Present
Based at KAIST; degree affiliation: Korea University.
Graduate Researcher | Advisor: Prof. Hyunwoo J. Kim

- Conducted research on long-video generation using 3D geometry and memory retrieval to maintain scene consistency over time.
- Conducted research on multimodal 3D perception, 3D scene and shape representations, efficient generative modeling, and visual representation learning for vision–language–action models.

BREIN Lab, Korea University — Jun 2020–Jun 2021
Undergraduate Researcher | Advisor: Prof. Joon-Kyung Seong

- Studied graph neural networks for resting-state fMRI (rs-fMRI) brain-network decomposition and salient functional connectivity.

RESEARCH PROJECTS

3D High-Level Scene Understanding and Situation Prediction for Urban Drones — 2023–Present
Research Lead | NRF Mid-Career Research Program

- Collected 25,671 synthetic image–point-cloud samples from simulated drone viewpoints in GTA5 and conducted object detection experiments for urban scene understanding.
- Currently conducting research on motion prediction, 3D reconstruction, and agentic automatic labeling.

Computer Vision for Breast Cancer Biomarker Development — 2023
Research Collaborator | Korea University Ansan Hospital

- Quantified extracellular matrix components and detected and counted immune cells in fluorescence microscopy images for breast cancer biomarker research. Related publication: [J1].

TECHNICAL SKILLS

Programming & tools: Python, PyTorch, MATLAB, diffusers, PyTorch3D, Linux.

HONORS & AWARDS

Minister's Award, 3rd Place — 2021
AI Grand Challenge for Disaster Response Drones, Ministry of Science and ICT, Republic of Korea

PUBLICATIONS & PREPRINTS

*Equal contribution.

Conference Papers

[C5] Retrieve What's Missing: Coverage-Maximizing Retrieval for Consistent Long Video Generation
Minseok Joo, Dogyun Park, Taehoon Lee, Kyujin Lee, Hyunwoo J. Kim
Advances in Neural Information Processing Systems (NeurIPS), 2026. [Paper] [Project]

[C4] F^{4}Splat: Feed-Forward Predictive Densification for Feed-Forward 3D Gaussian Splatting
Injae Kim, Chaehyeon Kim, Minseong Bae, **Minseok Joo**, Hyunwoo J. Kim
European Conference on Computer Vision (ECCV), 2026. [Paper] [Project] [Code]

[C3] Blockwise Flow Matching: Improving Flow Matching Models For Efficient High-Quality Generation
Dogyun Park, Taehoon Lee, **Minseok Joo**, Hyunwoo J. Kim
Advances in Neural Information Processing Systems (NeurIPS), 2025. [Paper] [Code]

[C2] Generative Subgraph Retrieval for Knowledge Graph–Grounded Dialog Generation
Jinyoung Park, **Minseok Joo**, Joo-Kyung Kim, Hyunwoo J. Kim
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024. [Paper] [Code]

[C1] Semantic-Aware Implicit Template Learning via Part Deformation Consistency
Sihyeon Kim, **Minseok Joo**, Jaewon Lee, Juyeon Ko, Juhan Cha, Hyunwoo J. Kim
IEEE/CVF International Conference on Computer Vision (ICCV), 2023. [Paper] [Code]

Journal Articles

[J1] Optimized prediction of breast cancer tumor microenvironment using MRI-based intratumoral and peritumoral radiomics: a prospective study
Eun Sil Kim, Sungwon Ham, Bo Kyoung Seo, Ji Young Lee, Woong Sun, Minkyu Jeon, **Minseok Joo**, Seonghoon Park, Shuncong Wang, Boram Lee, Hye Yoon Lee, Min Sun Bae, Kyu Ran Cho, Ok Hee Woo, Sung Eun Song, Soo-Yeon Kim
Frontiers in Oncology, 15:1654508, 2025. [Paper]
In collaboration with Korea University Ansan Hospital.

Preprints

[P4] RLSplat: Resource-Linear Budget Control for Feed-forward 3D Gaussian Splatting
Injae Kim, **Minseok Joo**, Jihwan Park, Sehyung Kim, Minseong Bae, Hyunwoo J. Kim
arXiv:2610.27712, 2026. [Paper] [Project] [Code]

[P3] Retrieval-Augmented Skill Optimization via Cross-Harness Adaptation
Jaewon Chu, Ji Soo Lee, Jihwan Park, Dohwan Ko, Jeehye Na, Seunghun Lee, Taehoon Lee, Minseo Yoon, **Minseok Joo**, Yunyang Xiong, Hyunwoo J. Kim
arXiv:2609.38024, 2026. [Paper]

[P2] Mitigating State Aliasing in Vision-Language-Action Models via Inverse Dynamics Learning
Kyujin Lee, Injae Kim, Jihwan Park, Yejun Ju, **Minseok Joo**, Hyunwoo J. Kim
arXiv:2605.29577, 2026. [Paper]

[P1] Robust Multimodal 3D Object Detection via Modality-Agnostic Decoding and Proximity-based Modality Ensemble
Juhan Cha*, **Minseok Joo***, Jihwan Park, Sanghyeok Lee, Injae Kim, Hyunwoo J. Kim
arXiv:2407.19156, 2024. [Paper] [Code]

TEACHING & ACADEMIC SERVICE

Teaching Assistant, Korea University: Artificial Intelligence (Spring 2023, 2024); Deep Learning (Fall 2023, 2024).
Teaching Assistant, LG AI Workshop Jan 2023
Conference Reviewer: CVPR 2026; ICML 2026; ECCV 2026; NeurIPS 2026; ICLR 2027.